# CM3121 – Advanced Organic Chemistry

Dr. Zhu Ye

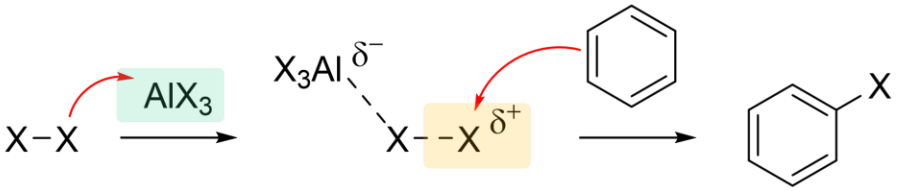
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Electrophilic Aromatic Substitution (Prat 2)

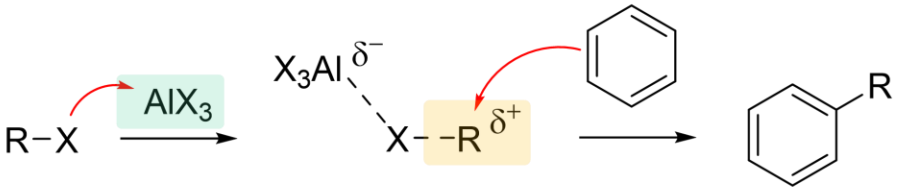
# Electrophilic Aromatic Substitution

**1. Bromination** or **chlorination**

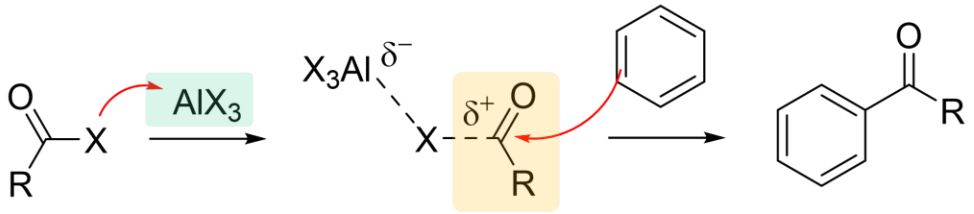



X = Br or Cl; $AlX_3$ can be other **Lewis acid** (e.g., $FeX_3$)

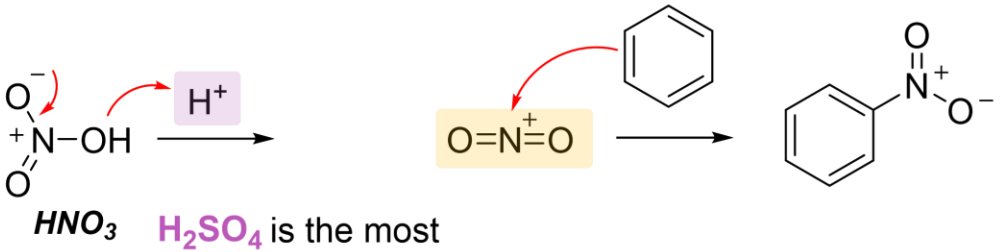
**2A. Friedel-Crafts alkylation**




**2B. Friedel-Crafts acylation**

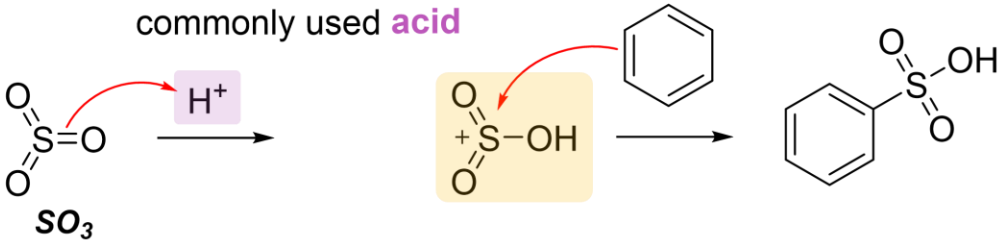



**3. Nitration**

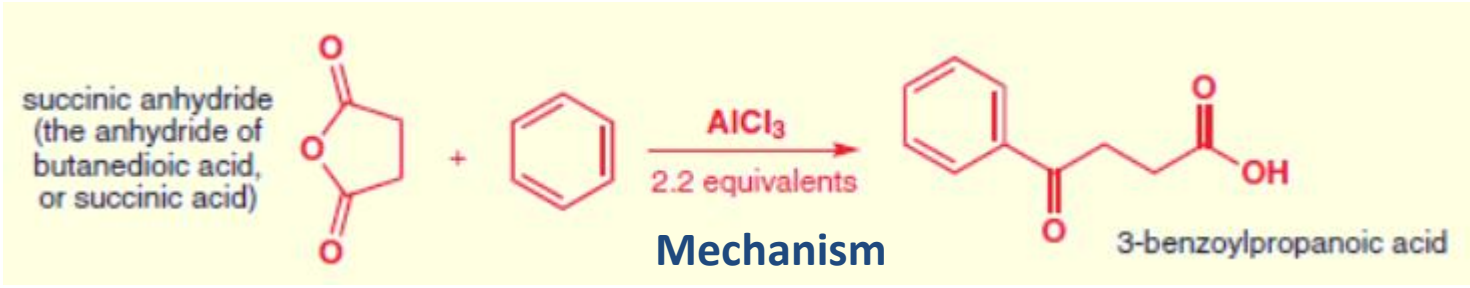


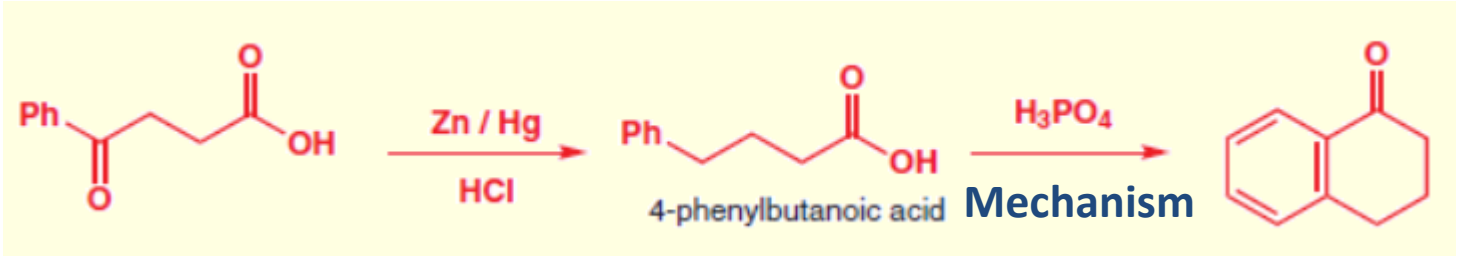

***$HNO_3$*** **$H_2SO_4$** is the most commonly used **acid**

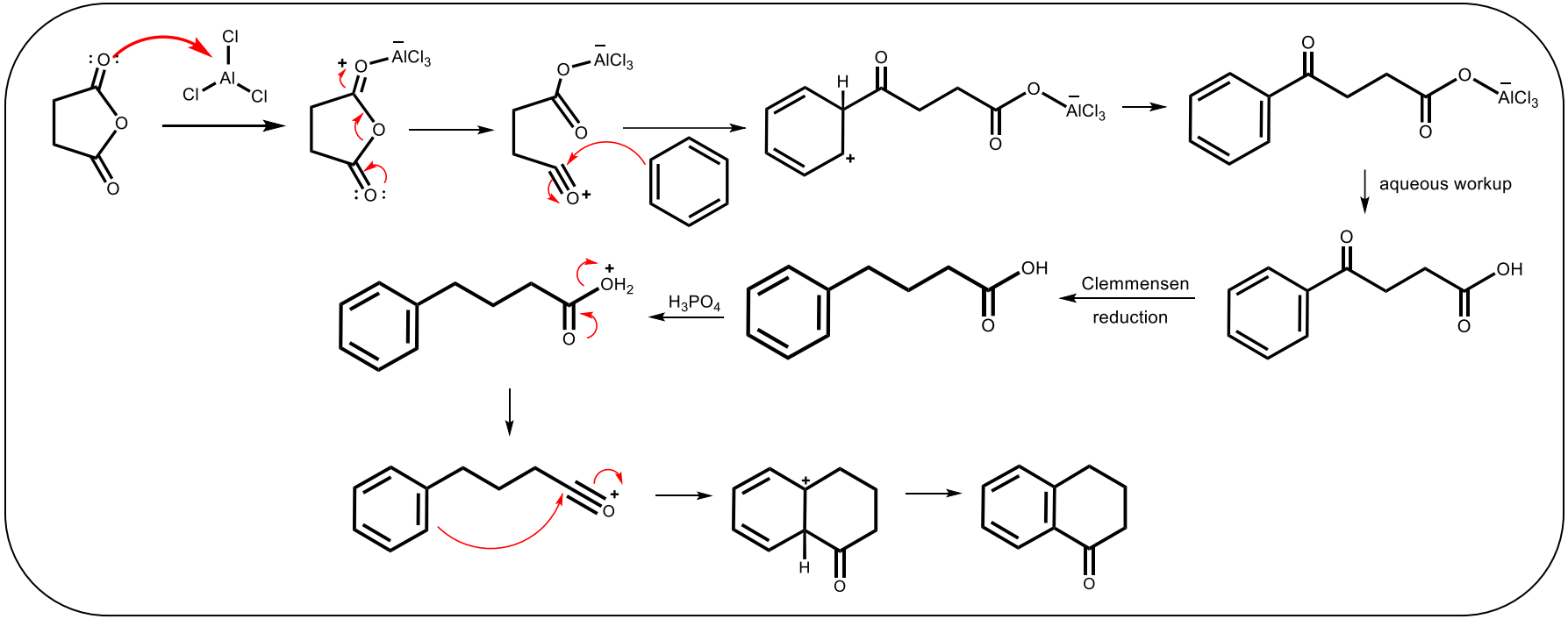
**4. Sulfonation**




***$SO_3$***

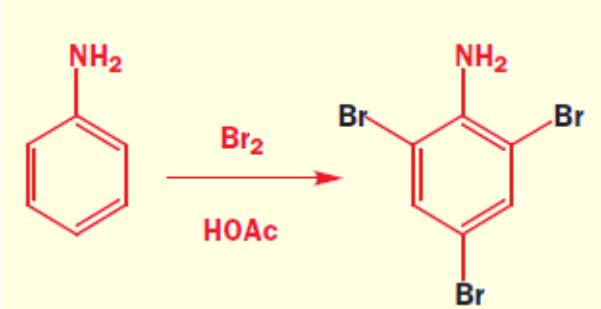

succinic anhydride
(the anhydride of
butanedioic acid,
or succinic acid)
AlCl3
2.2 equivalents
Mechanism
OH
3-benzoylpropanoic acid


Ph
OH
Zn / Hg
HCl
Ph
OH
4-phenylbutanoic acid
H3PO4
Mechanism

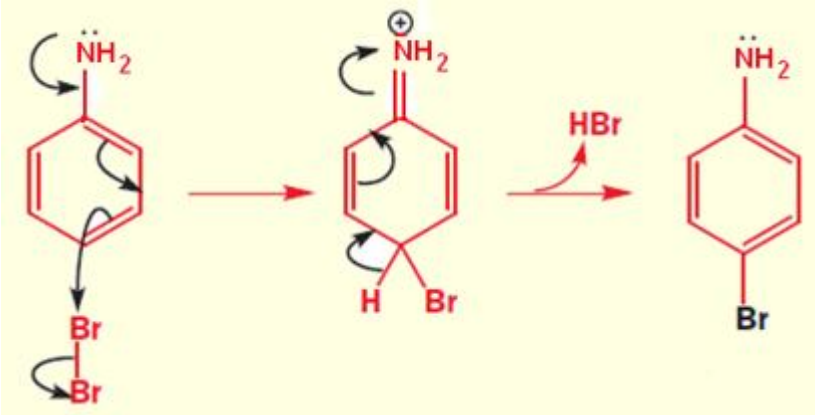

AlCl3
aqueous workup
Clemmensen
reduction
H3PO4
OH2
OH

# 8. Reactions of Anilines: A Nitrogen Lone Pair Activates the Arene Even More Strongly

Aniline (phenylamine) is even more reactive towards electrophiles than phenols, phenyl ethers, or phenoxide ions. Because nitrogen is less electronegative than oxygen

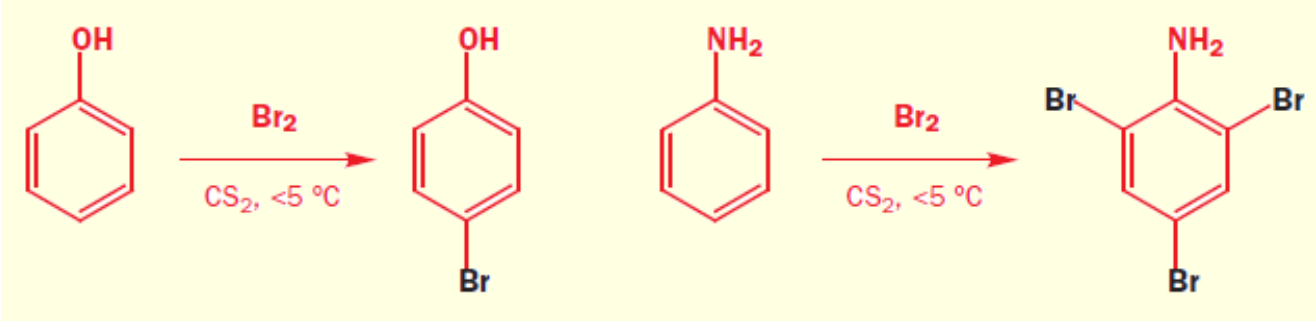



NOTE effect on orientation: $NH_2$ is ortho, para-directing



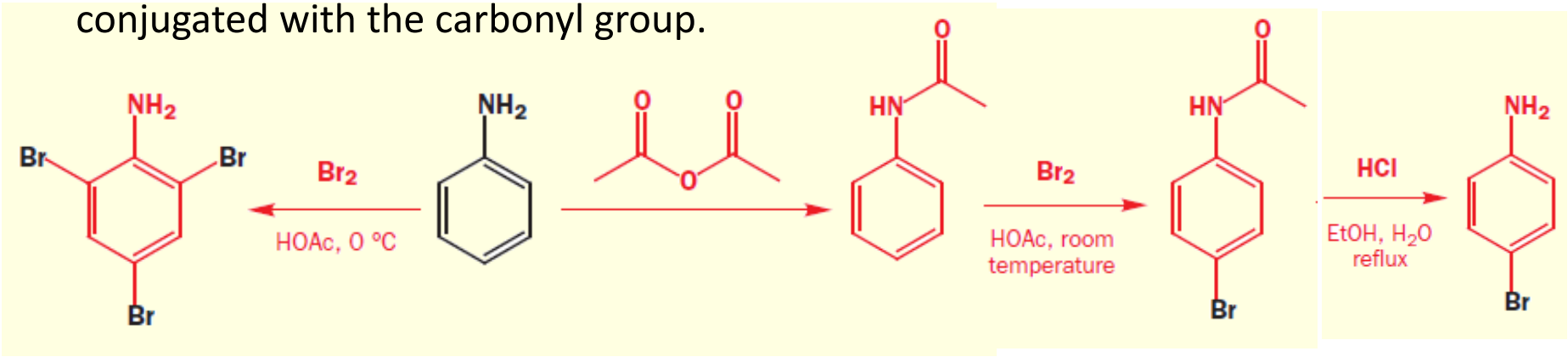

To put just one bromine atom on to the ring:

- for phenol, if bromine is added slowly to a solution of phenol in carbon disulfide and the temperature is kept below 5 °C, the main product is *para-bromophenol.*

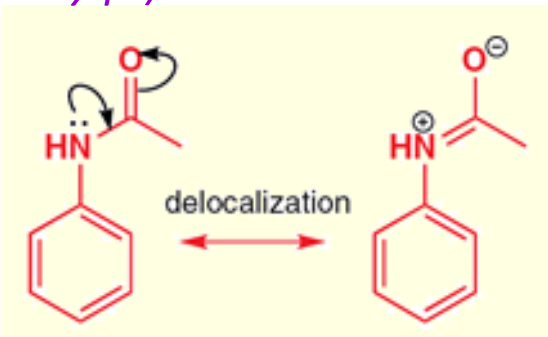


# How then could we prevent oversubstitution from occurring?

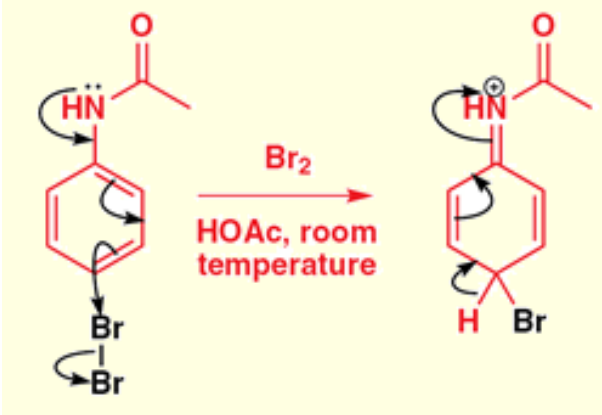
Recall: N atom in an amide is much less basic than a normal amine because it is conjugated with the carbonyl group.




Lone pair electrons on the N atom of the amide are conjugated with the C=O group and are also delocalized into the benzene ring, though more weakly than in the amine. Reaction still occurs in the *o- and p- positions (mainly p-) but it occurs once only.*




In general: 1st electrophilic substitution at para due to steric hindrance.



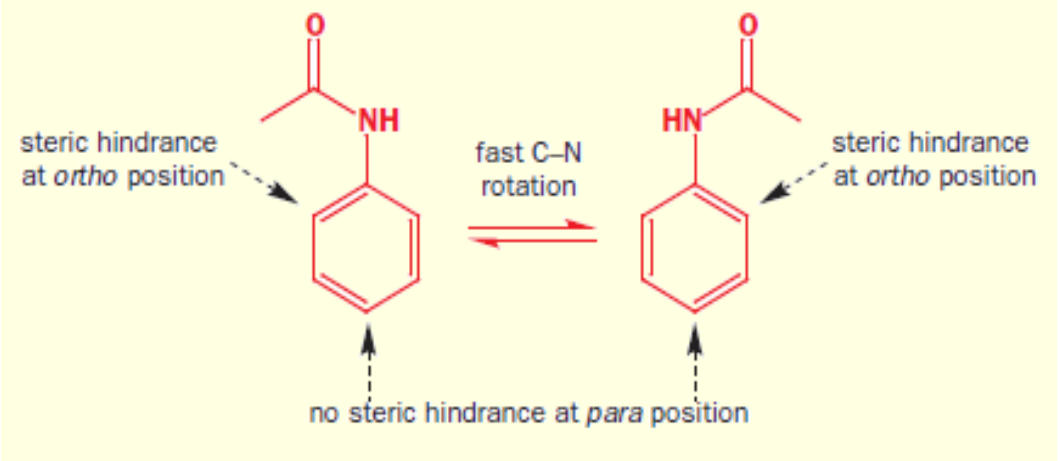
# Selectivity between *o- and p- positions is determined by steric hindrance*

Recall: Phenols and anilines react in the *o- and/or p- positions for electronic reasons*

When it comes to choosing between *o- and p- positions we need to consider* ***steric effects*** *as* well.

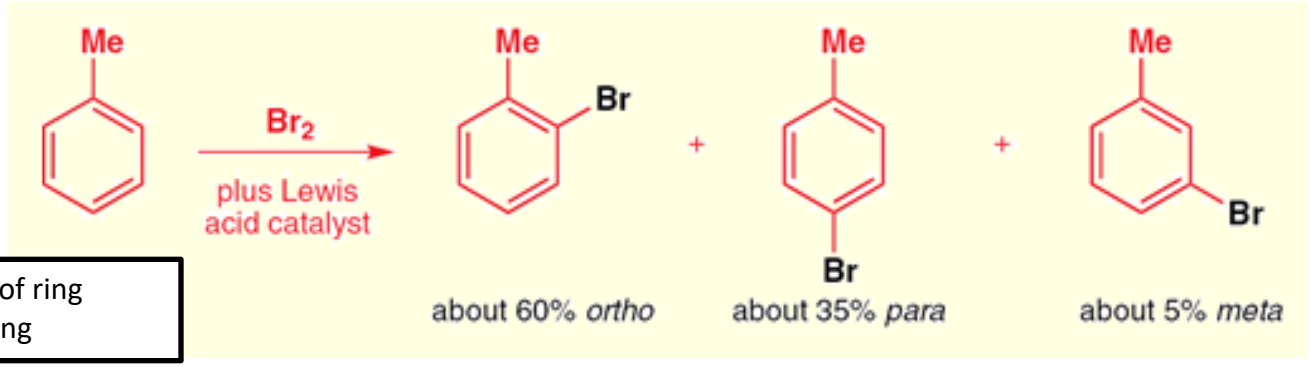
If the reactions occurred merely statistically, we should expect twice as much *o- as p- product* because there are two *o- positions. However, we should also expect more steric hindrance in o- substitution since the new substituent must sit closely beside the one already there.*

*With large* substituents, such as the amide, steric hindrance will be significant and it is not surprising that we get more *p- product.*



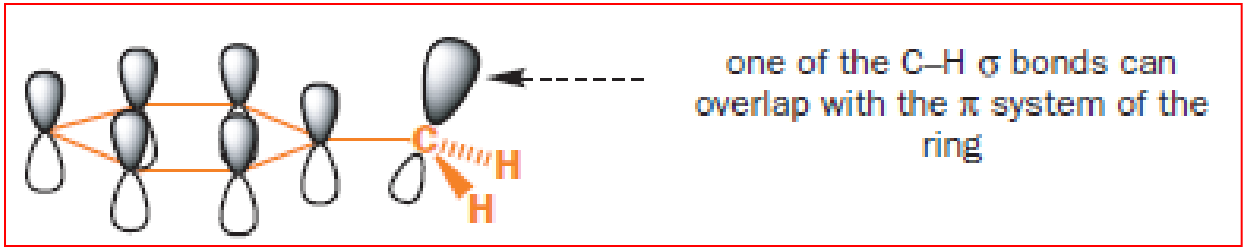
# 8. Alkyl benzenes also react at the *o- and p-positions:* $\sigma$ *donor* substituents

The methyl group also directs electrophiles mostly into the *o- and p- positions. These two* observations together suggest that alkyl groups may also increase the electron density in the $\pi$ system of the benzene ring, specifically in the *o- and p-positions, rather like a weakened version of an* OR group.

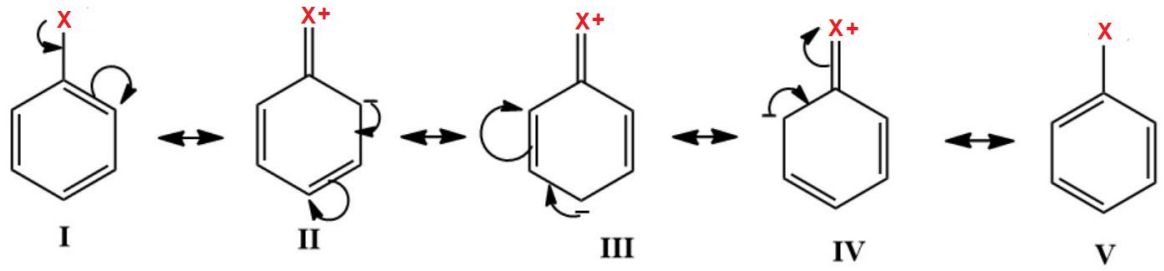



- Me increases reactivity of ring
- Me is ortho,para-directing

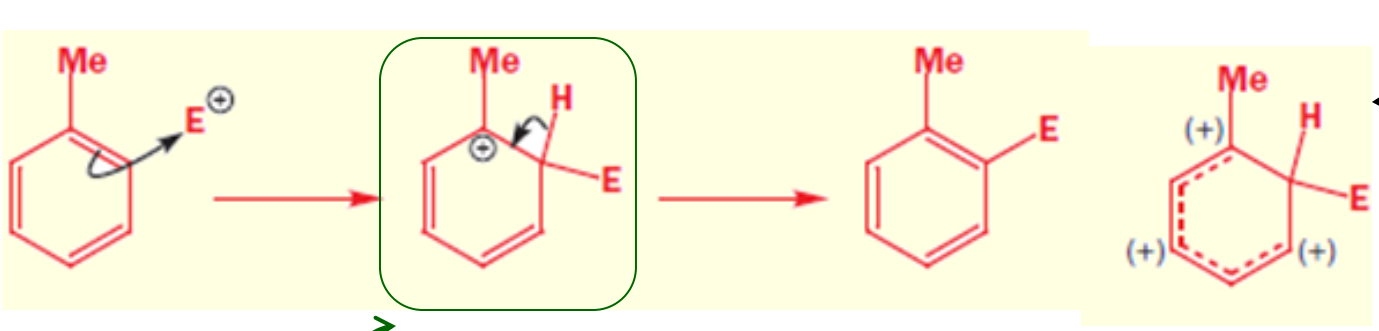
The methyl group donates electrons weakly by conjugation – **$\sigma$ conjugation**




An alkyl group behaves like an electron-donating group

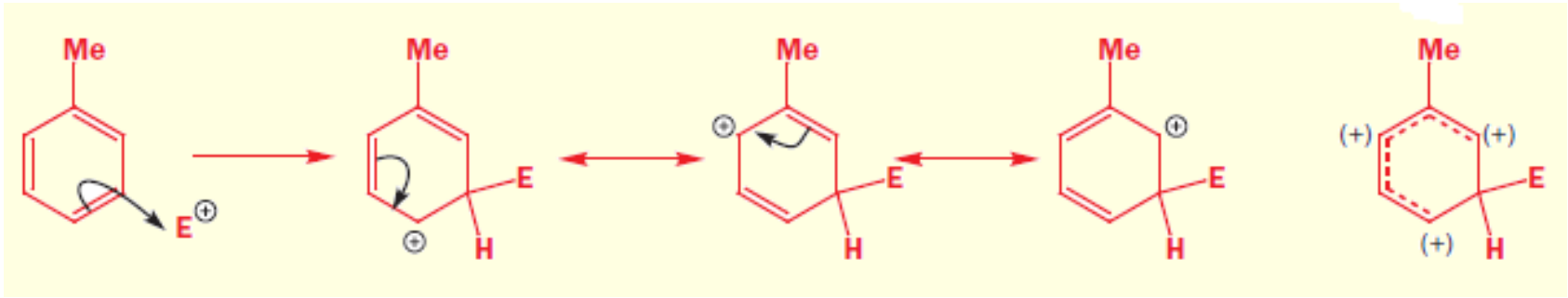


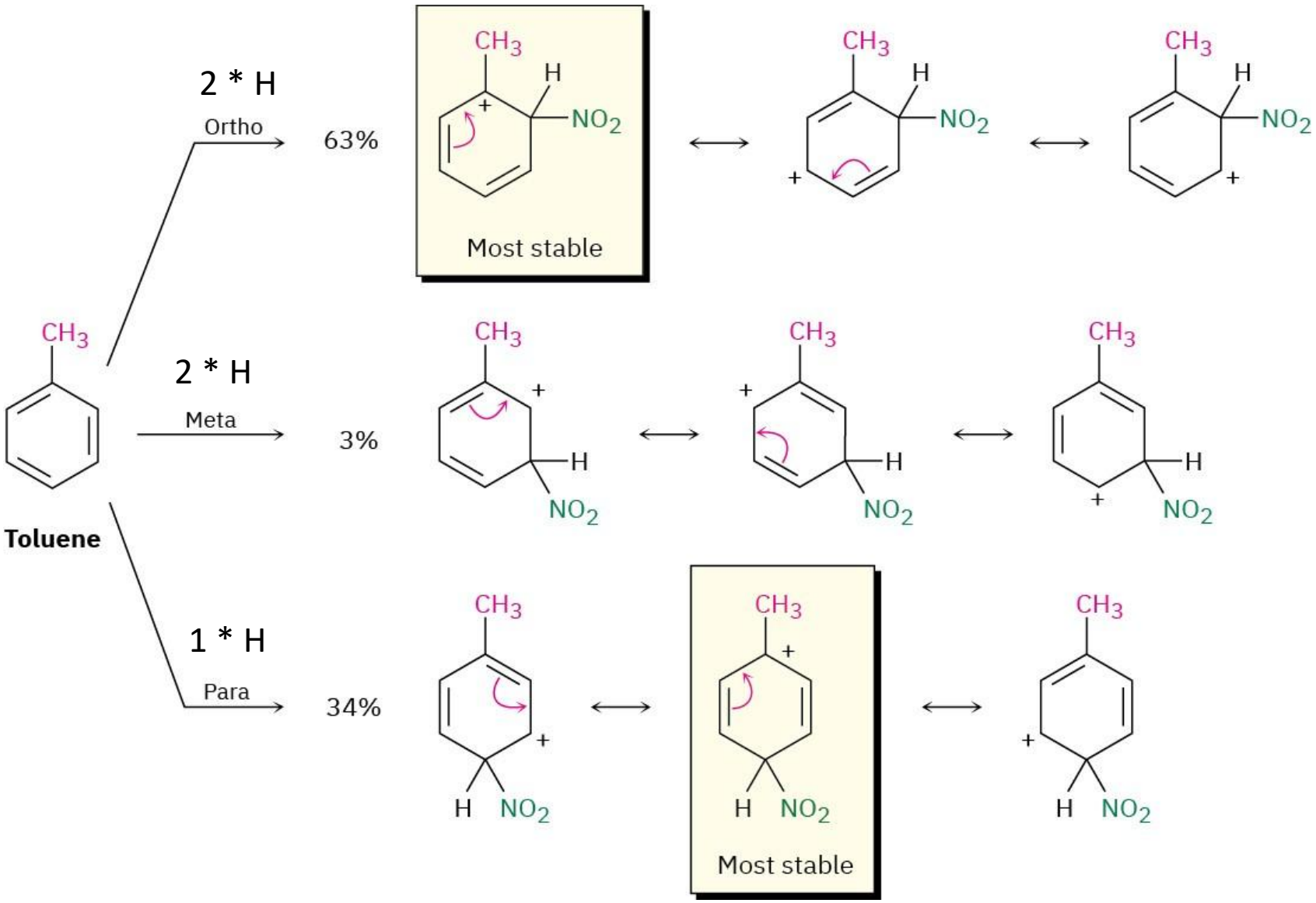

positive charge in the intermediate is delocalized over three carbons




This carbon is tertiary, making the cation there more stable. This condition is fulfilled if toluene is attacked at the *o- or p- positions.*

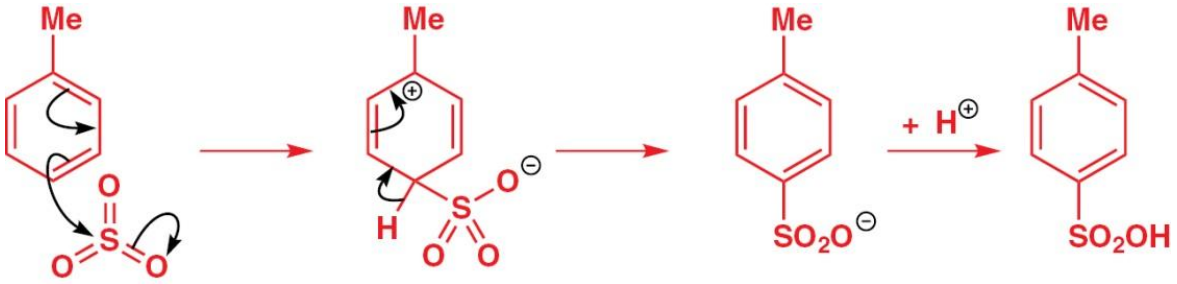
In Contrast, the Me can not stabilize the cation *if it is attacked at the m- position*

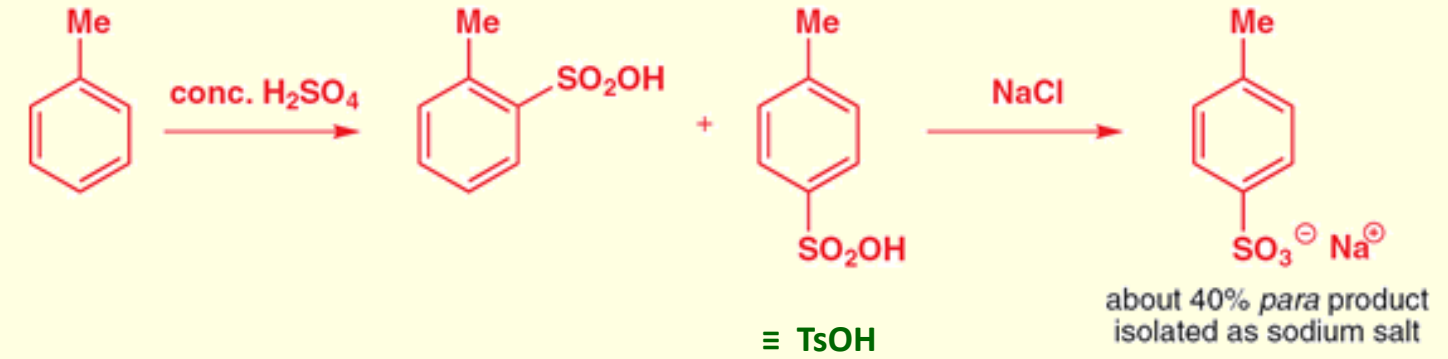



2 * H
Ortho
63%
Most stable
2 * H
Meta
3%
Toluene
1 * H
Para
34%
Most stable

# Sulfonation of toluene

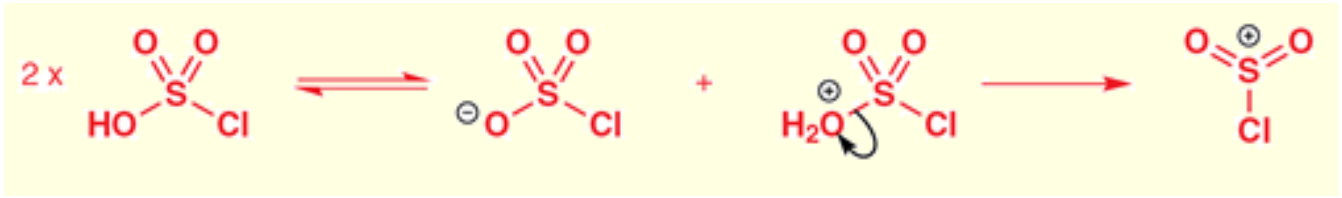
Direct sulfonation of toluene with concentrated sulfuric acid gives a mixture of *ortho- and para- toluenesulfonic* acids

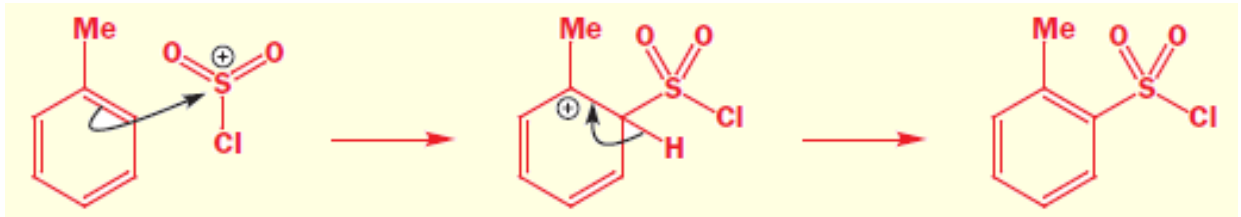


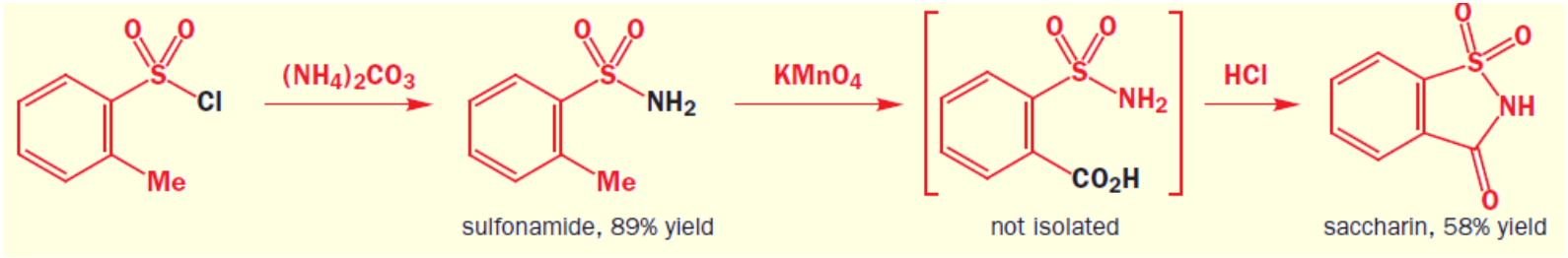



# Application of sulfonation of toluene

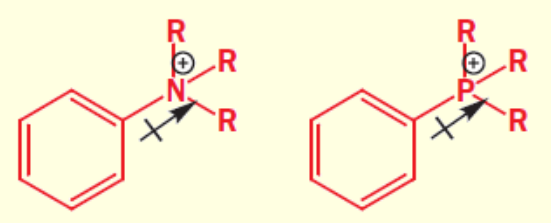


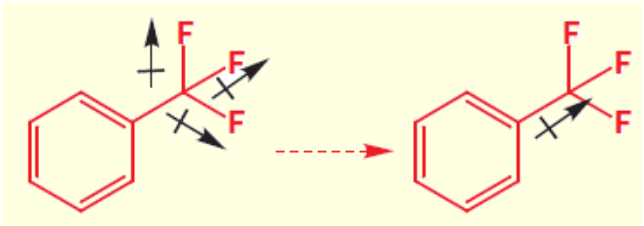



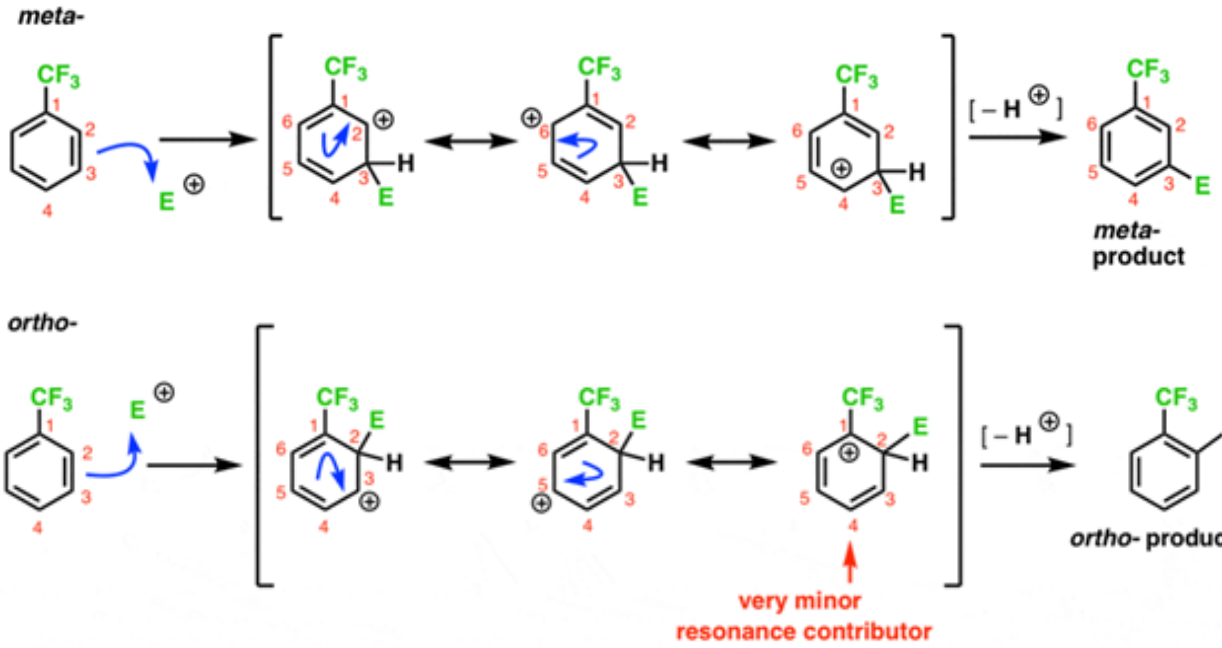
## 9. Electron-withdrawing substituents give *meta products*




- Ar–$N^+$ and Ar–$P^+$ bonds are polarized towards the positively charged heteroatom;
- N is tetrahedral and no longer has lone pairs to donate



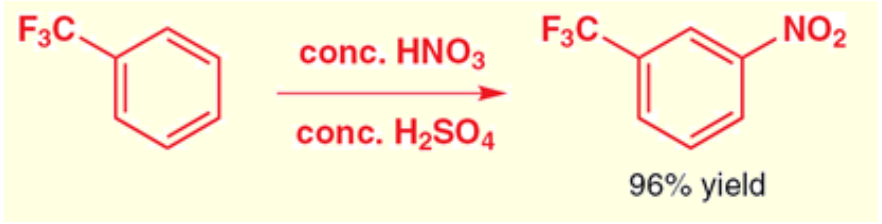

Ar–C bond in Ar–$CF_3$ is polarized towards the $CF_3$ group because of the 3 electronegative fluorine atoms polarizing the C–F bonds so much that the Ar–C bond is polarized too

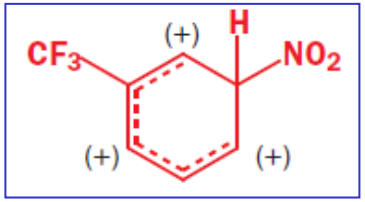


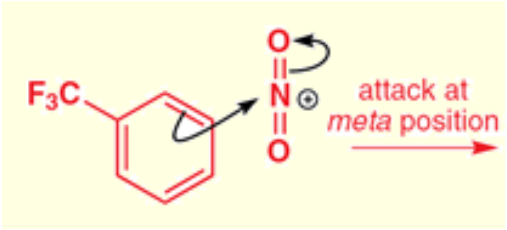

Electron-withdrawing substituents direct electrophiles to the *meta position and reduce reactivity ie* **deactivating** and **meta-directing**

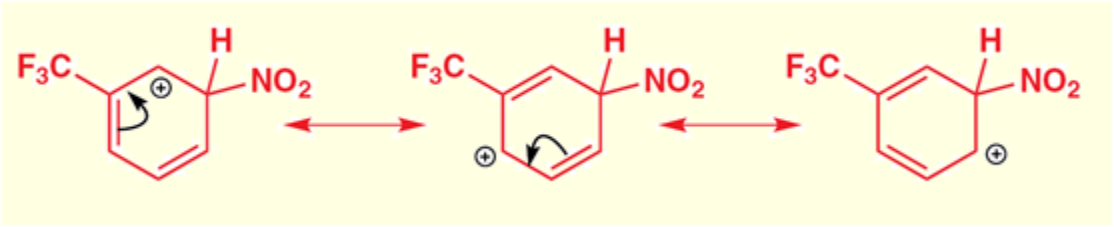
# Applications

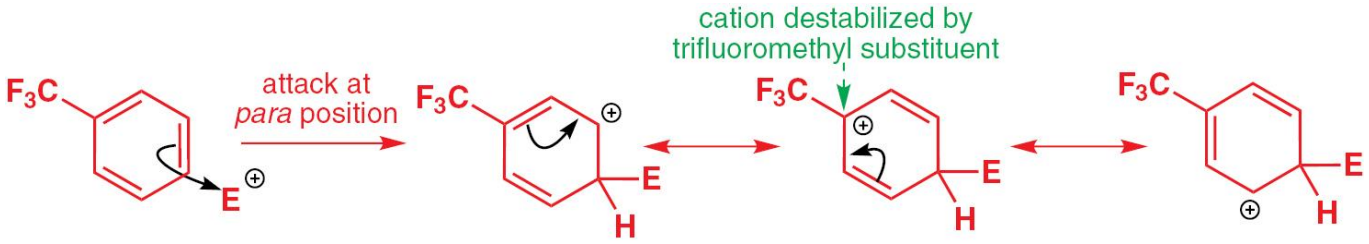


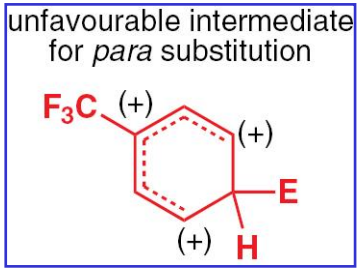



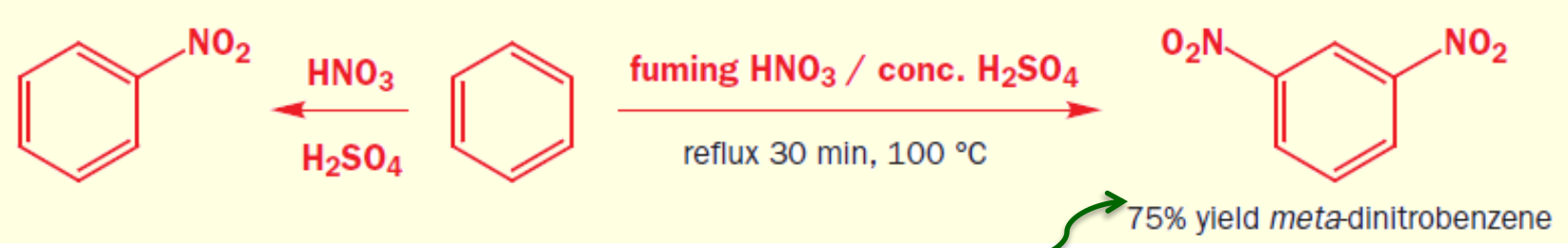



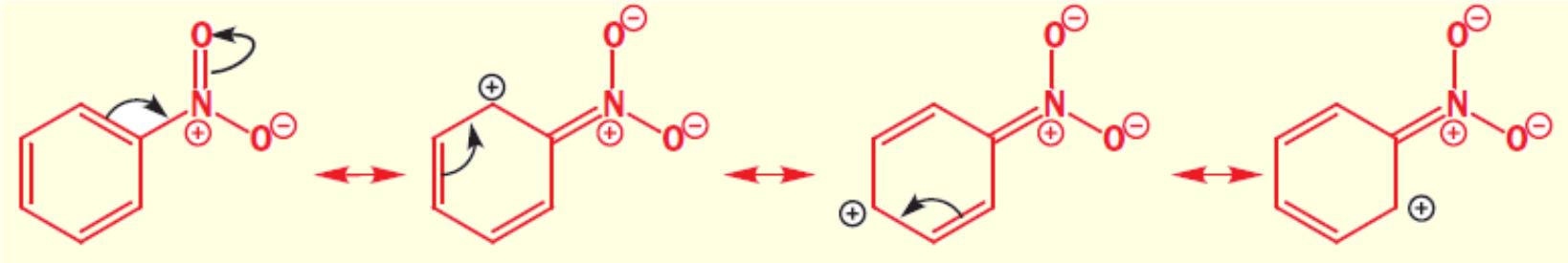



# Some substituents withdraw electrons by conjugation



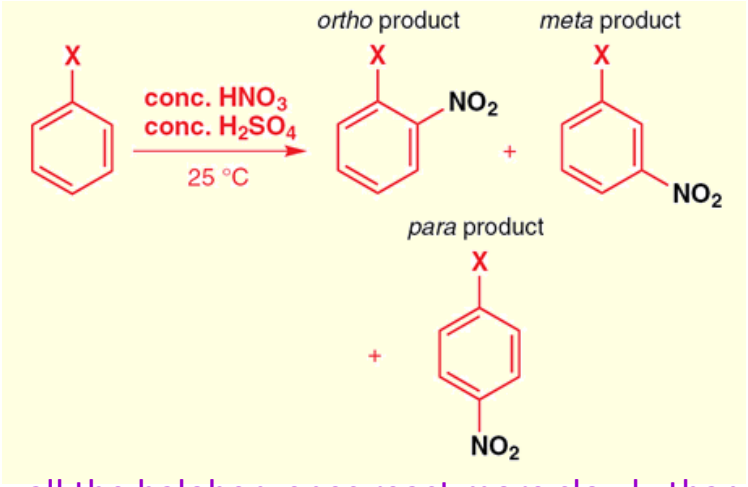

second nitro group is introduced *meta to the first, ie,* the nitro group is deactivating and *meta-directing*

$NO_2$ is conjugated with the π system of the benzene ring and is strongly electron-*withdrawing.* It withdraws electrons specifically from the *o- and p-positions.*



$NO_2$ withdraws electron density from the π system of the ring thereby making the ring less reactive towards an electrophile. Since more electron density is removed from the *o*- and *p*- positions, the least electron-deficient position is the *m*-position.

# 10. Halogens both withdraw and donate electrons




| Compound | Products formed (%) *ortho* | *meta* | *para* | Nitration rate (relative to benzene) |
|---|---|---|---|---|
| PhF | 13 | 0.6 | 86 | 0.18 |
| PhCl | 35 | 0.9 | 64 | 0.064 |
| PhBr | 43 | 0.9 | 56 | 0.060 |
| PhI | 45 | 1.3 | 54 | 0.12 |

- all the halobenzenes react more slowly than benzene

  ⇨ electron-withdrawal by the electronegative halogen deactivates the ring towards attack

- halogens are *o*-, *p*- directing

  ⇨ there are two opposing effects: electron donation by conjugation and electron withdrawal by induction

- lone pair on halogen may conjugate with the ring but does so less well than phenol or aniline

  ⇨ Cl, Br and I are larger in size than C, thus 2p orbital of C overlap poorly with the bigger p orbitals from halogen; fluorine has the right size but is very electronegative

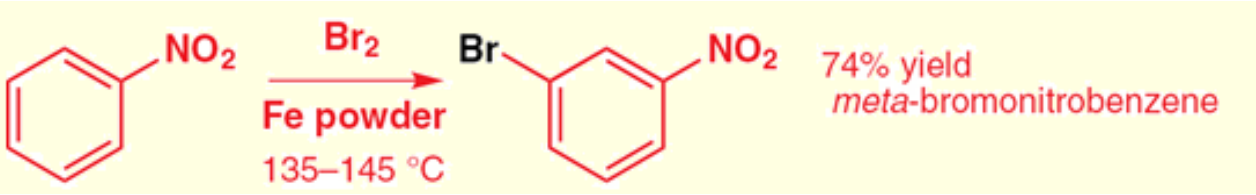
- increase in o- product from fluoro- to iodobenzene

  ⇨ more electronegative atoms have greater inductive effect and withdraws electrons mostly from *o*-position, hence lessening their reactivity

# Summary for the electronic effects

| Electronic effect | Example | Activation | Direction |
|---|---|---|---|
| donation by conjugation | $-NR_2$, $-OR$ | very activating | *ortho*, *para* only |
| donation by inductive effect | alkyl | activating | mostly *ortho*, *para* but some *meta* |
| donation by conjugation *and* withdrawal by inductive effect | F, Cl, Br, I | deactivating | *ortho* and (mostly) *para* |
| withdrawal by inductive effect | $-CF_3$, $-NR_2^+$ | deactivating | *meta* only |
| withdrawal by conjugation | $-NO_2$, $-CN$, $-COR$, $-SO_3R$ | very deactivating | *meta* only |

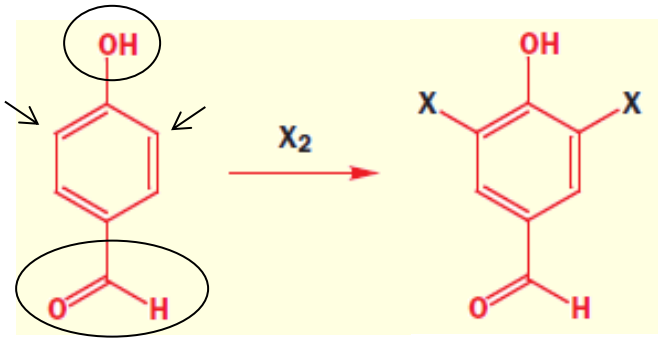
Reactions such as nitration, sulfonation, and Friedel–Crafts acylation add a very deactivating substituent. They stop cleanly after a single substitution unless there is harsh reaction condition.

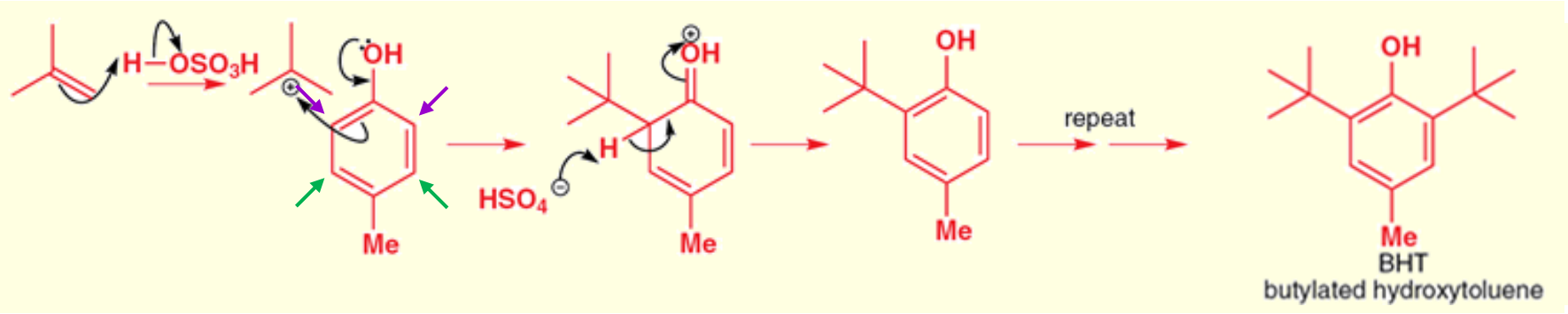


# 11. Two or more substituents may cooperate or compete

the directing effects of two or more substituents can work with or against one another

- aldehyde directs *meta*
- OH group directs *ortho*
- *so they both direct to the same position.*

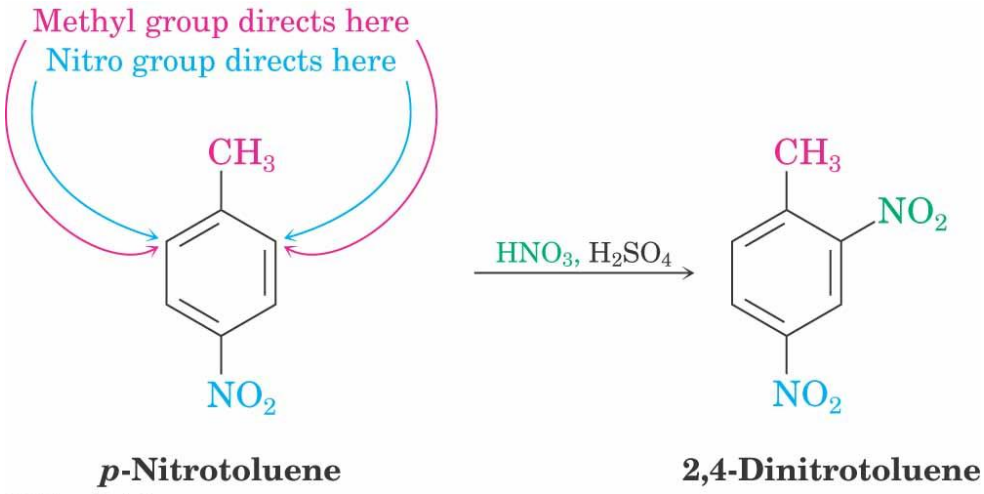


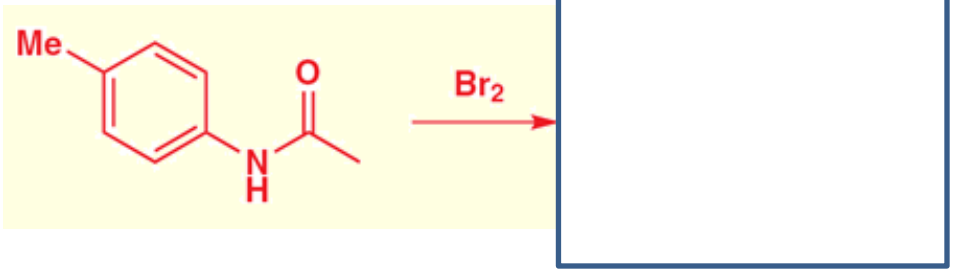

- OH and Me groups are *o*-, *p*- directing
- OH and Me competes in directing effect; *OH group* is much more powerfully directing than the methyl group because it provides an extra pair of electrons, so it 'wins' and directs the electrophile (a *t-butyl cation) ortho to itself*

# Additivity of Effects

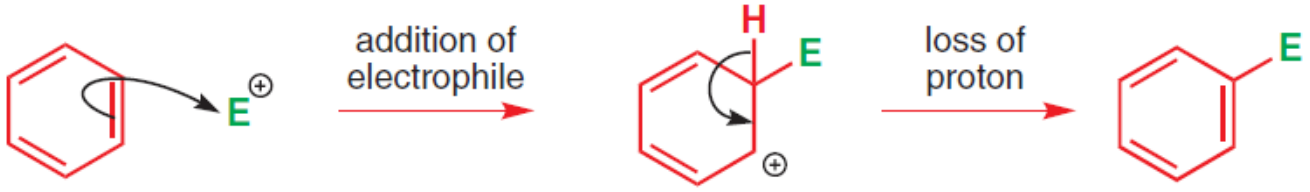
- If the directing effects of the two groups are the same, the result is additive

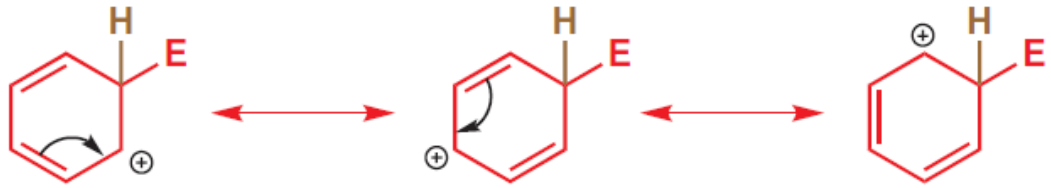


# Additivity of Effects

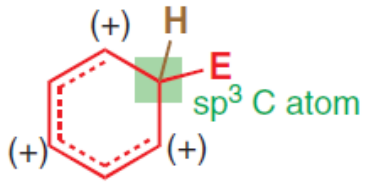


# 12. NMR as a predictive tool



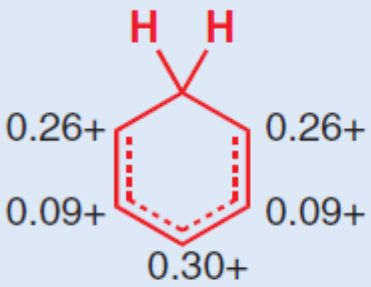




the brown H is drawn in to emphasize the non-aromaticity of this delocalized cation



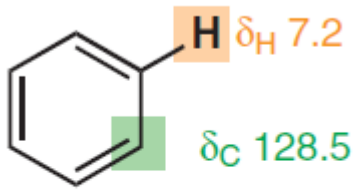

cyclohexadienyl cation



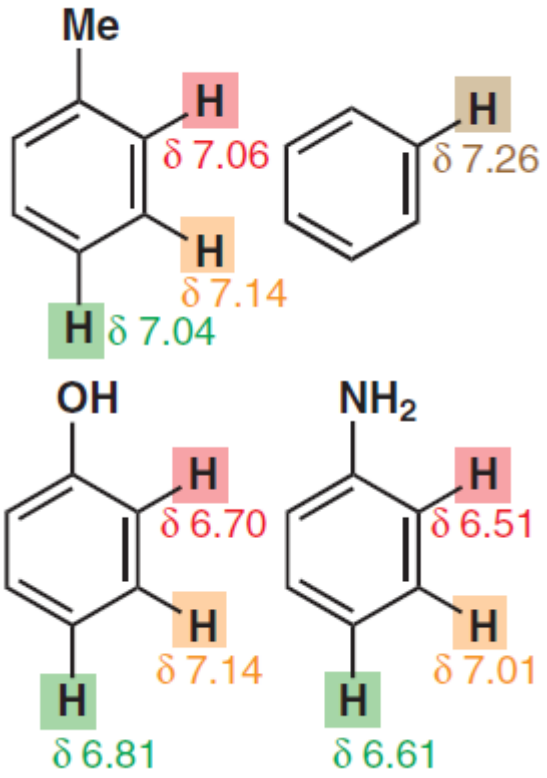

charge distribution

| position | $\delta_H$ | $\delta_C$ |
|---|---|---|
| 1 | 5.6 | 52.2 |
| 2,6 | 9.7 | 186.6 |
| 3,5 | 8.6 | 136.9 |
| 4 | 9.3 | 178.1 |
| benzene (for comparison) | 7.33 | 129.7 |




NMR data

$^1$H NMR shifts in

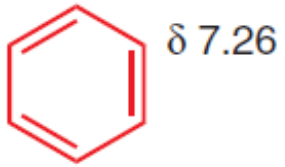




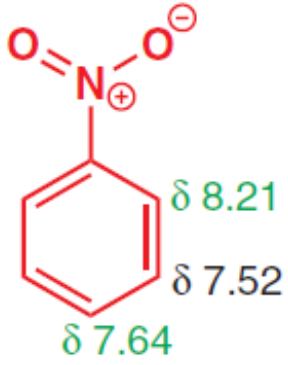


| Compound | R | Relative rate of bromination |
|---|---|---|
| benzene | H | 1 |
| methoxybenzene (anisole) | OMe | $10^9$ |
| *N,N*-dimethylaniline | $NMe_2$ | $10^{14}$ |

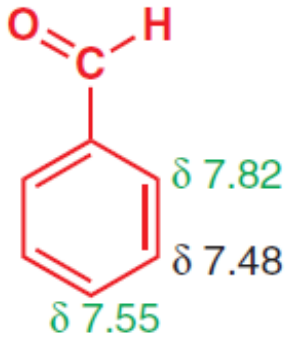
$^1$H NMR chemical shifts



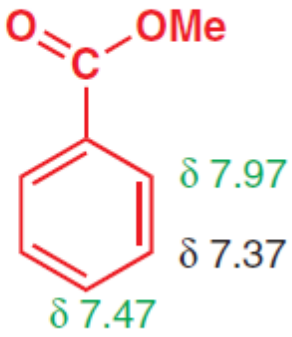

nitrobenzene



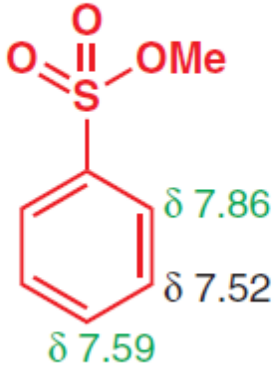

benzaldehyde



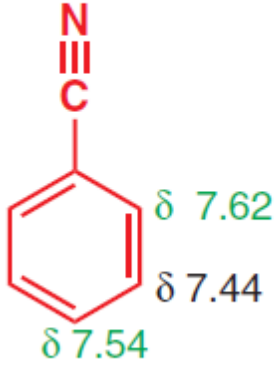

methyl benzoate



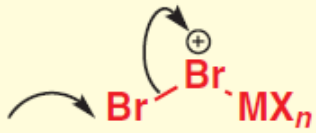
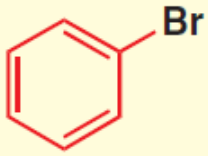
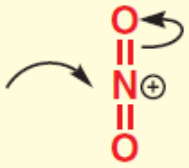
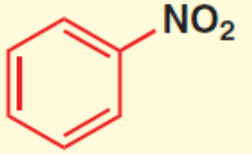
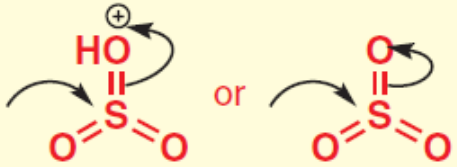
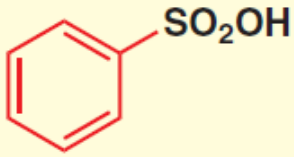
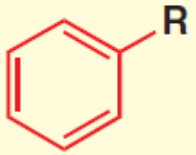
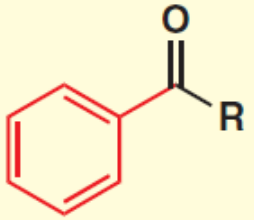

methyl benzenesulfonate



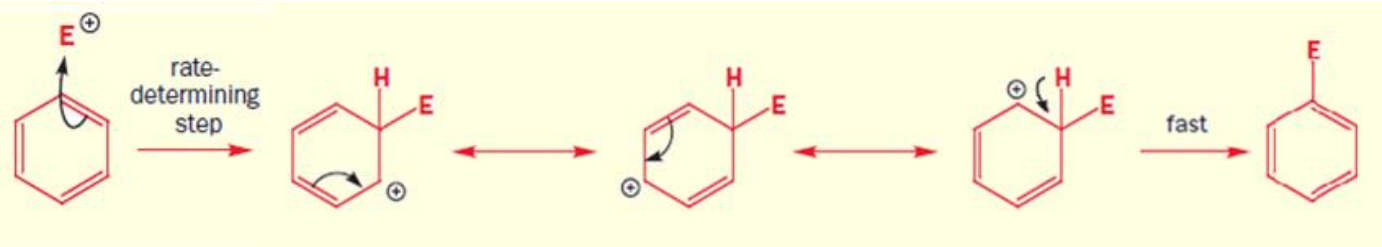

benzonitrile



# Summary

| Reaction | Reagents | Electrophile | Product |
|---|---|---|---|
| bromination | $Br_2$ and Lewis acid, e.g. $AlCl_3$, $FeBr_3$, Fe powder |  |  |
| nitration | $HNO_3 + H_2SO_4$ |  |  |
| sulfonation | concentrated $H_2SO_4$ or $H_2SO_4 + SO_3$ (oleum) |  |  |
| Friedel–Crafts alkylation | RX + Lewis acid usually $AlCl_3$ |  |  |
| Friedel–Crafts acylation | RCOCl + Lewis acid usually $AlCl_3$ |  |  |

# Mechanism




# Regioselectivity of Electrophilic Aromatic Substitution

